Principles of communication systems

EET3202, CUNY City Tech, Fall 2023

Homework #04 Solutions

S. M. Farzaneh

Problem 1

a) $v(t) = 2 = 2\cos(2\pi(0))$

b) $v(t) = -3 = -3\cos(2\pi(0)) = 3\cos(2\pi(0) + \pi)$

c) $v(t) = \sin(2\pi t) = \cos(2\pi(1)t - \pi/2)$

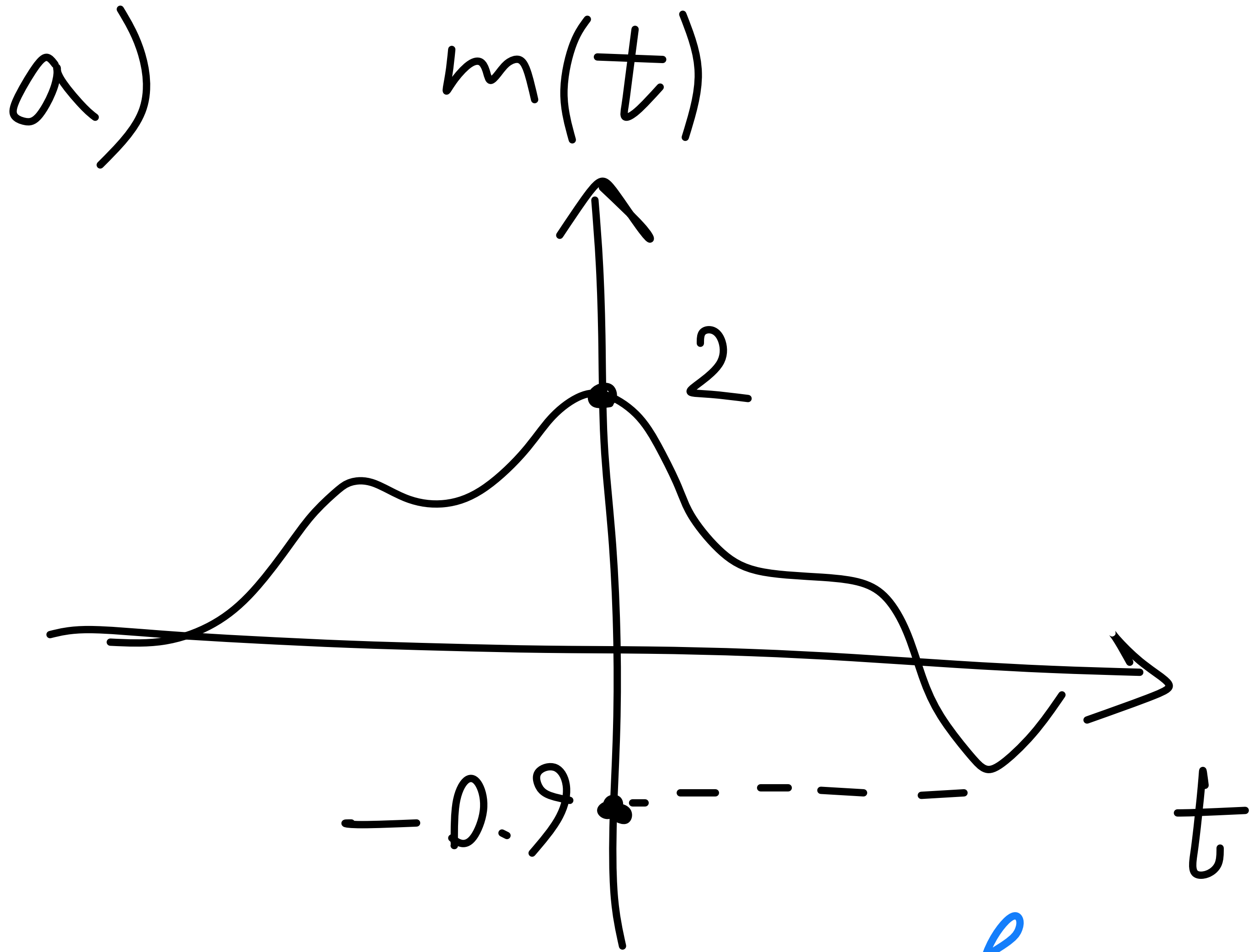
d) $v(t) = -\sin(2\pi t) = -\cos(2\pi(1)t - \pi/2) = \cos(2\pi(1)t + \pi/2)$

e) $v(t) = -0.5\cos(120\pi t + 45^\circ) = -0.5\cos(2\pi(60)t + \pi/4)$

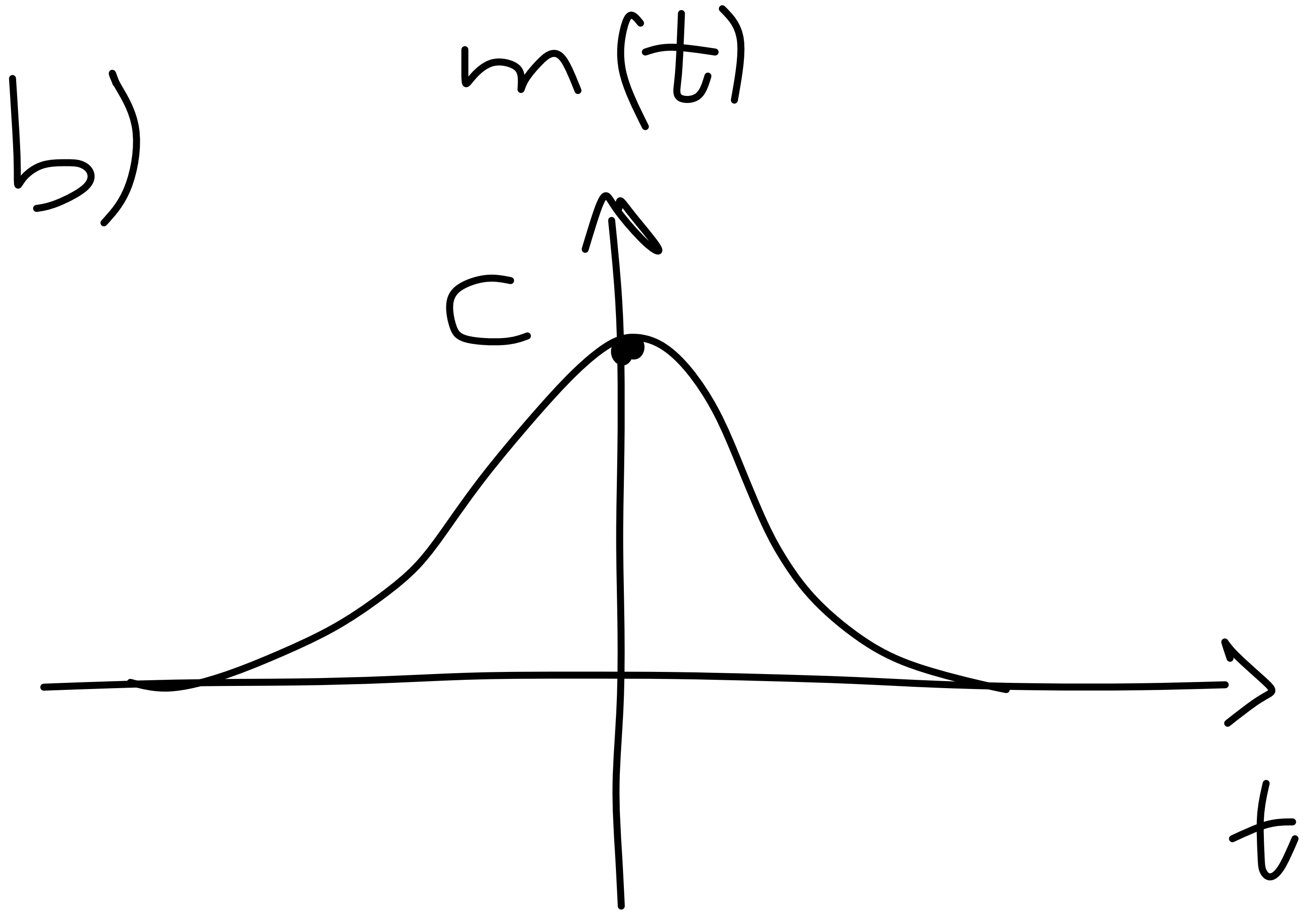
$= 0.5\cos\left(2\pi(60)t + \frac{5\pi}{4}\right)$

Problem 2

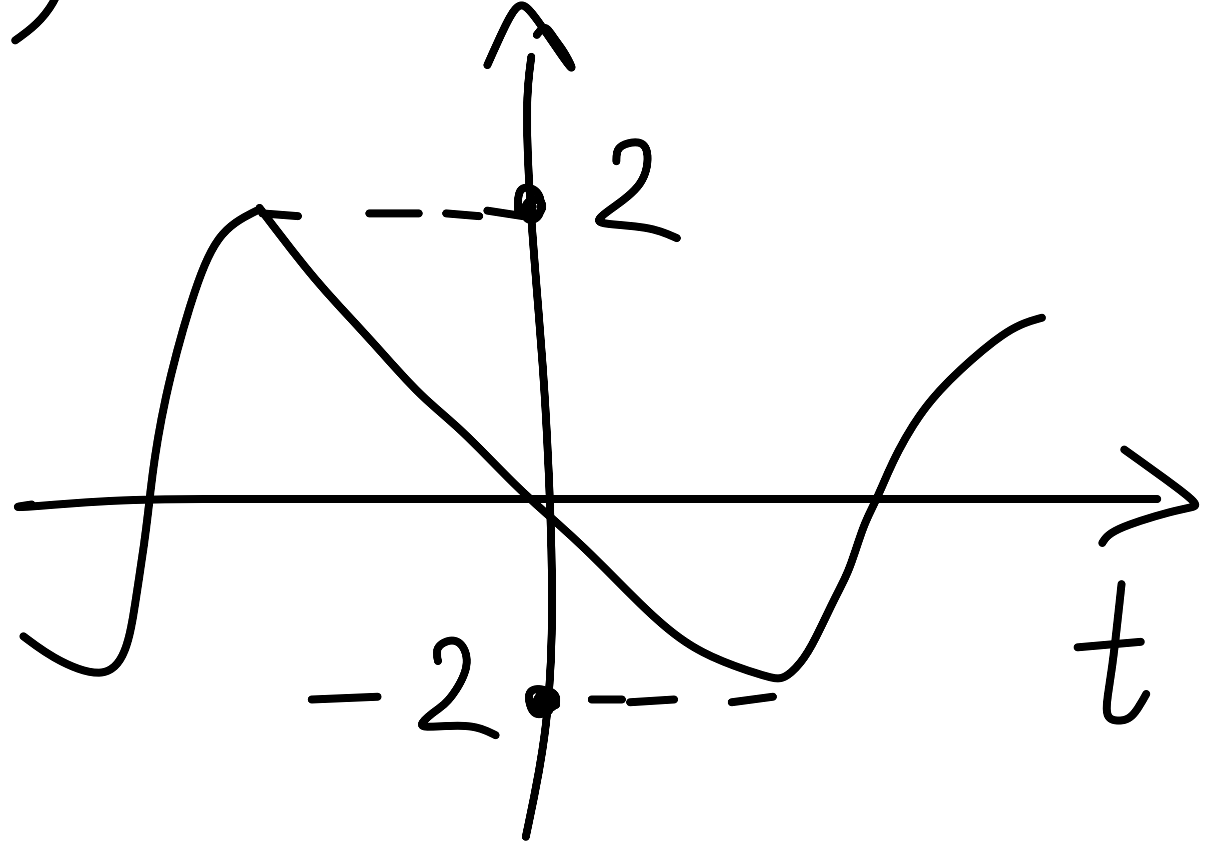
a)



b)



c)

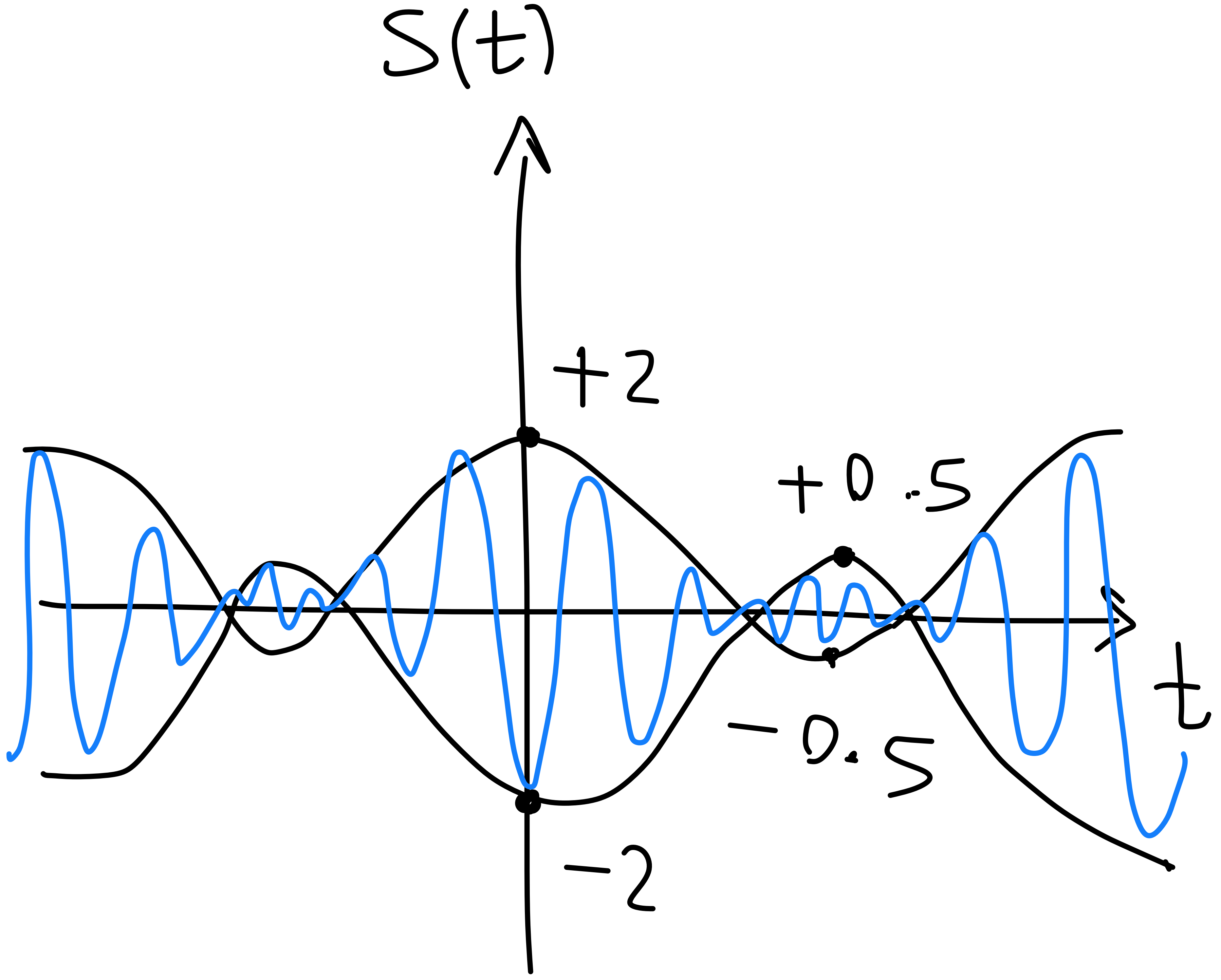


$$\mu = \frac{\text{range of } m(t)}{\text{DC voltage} + \text{average of } m(t)} = \frac{m_{max} - m_{min}}{A + \frac{m_{max}+m_{min}}{2}}$$

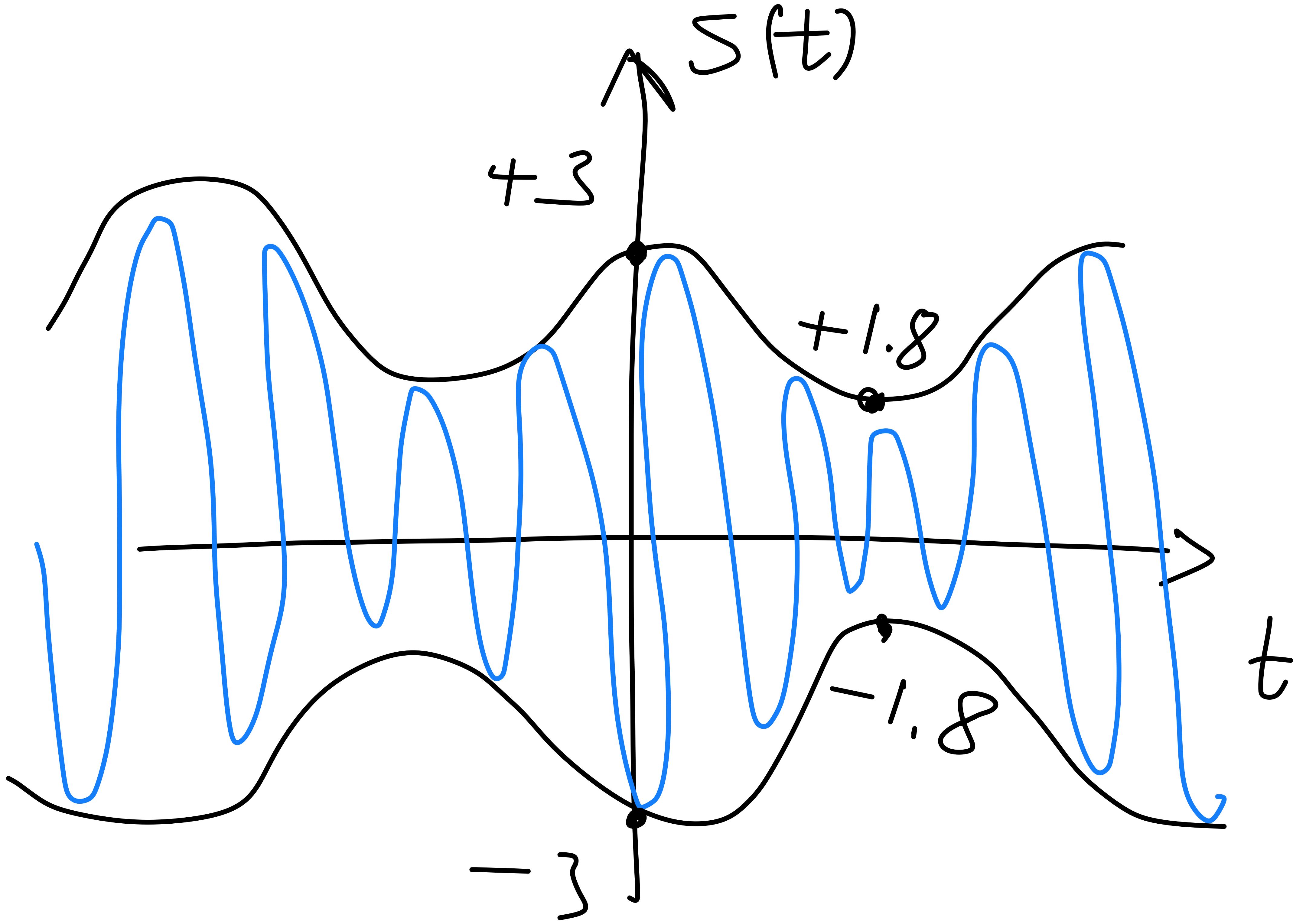
$$A=1 \Rightarrow \mu_a = \frac{2.9}{1+\frac{1.1}{2}} = \frac{2.9}{1.55}, \quad \mu_b = \frac{c}{1+\frac{c}{2}} = \frac{2c}{2+c}, \quad \mu_c = \frac{4}{1+0} = 4$$

Problem 3

a)



$$\mu = \frac{2-(-0.5)}{2+(-0.5)} = \frac{2.5}{1.5}$$



$$\mu = \frac{3-(1.8)}{3+(1.8)} = \frac{1.2}{4.8} = \frac{1}{4}$$

Problem 4

$$v(t) = A\cos(w_c t) + B\cos(w_m t)\cos(w_c t) \qquad T_c = \frac{1}{f_c} = \frac{2\pi}{w_c}$$

$$P_c = \frac{1}{T_c}\int_{-T_c/2}^{T_c/2} |A\cos(w_c t)|^2 dt = \frac{A^2}{T_c}\int_{-\frac{T_c}{2}}^{\frac{T_c}{2}} \left(\frac{1+\cos(2w_c t)}{2}\right) dt = \frac{A^2}{T_c}\cdot\frac{T_c}{2} = \frac{A^2}{2}$$

side band: $B\cos(w_m t)\cos(w_c t) = \frac{B}{2}\left(\cos((w_c+w_m)t) + \cos((w_c-w_m)t)\right)$

$$P_s = \frac{(B/2)^2}{2} + \frac{(B/2)^2}{2} = \frac{B^2}{4}, \quad \eta = \frac{P_s}{P_c+P_s}$$

a) $P_c = 2$, $P_s = 0.81$

$$\eta = \frac{0.81}{2.81} = 29\%$$

b) $P_c = 0.125$, $P_s = 1$

$$\eta = \frac{1}{1.125} = 89\%$$